Supplementary Material: Sparse Feature Selection Outweighs Deep Learning Capacity for Out-of-Region Transfer in Crop Classification

Michael L. Mann, Sairam Venkatachalam, Disha Kacha, Devarsh Sheth, Ryan Engstrom, Amir Jafari

This document collects supporting material for the main paper. Section S-A details the dataset, feature set, and model definitions; Section S-B defines the metrics; Section S-C reports training cost; Section S-D presents the ablation studies; Section S-E maps the spatial distribution of holdout errors; Section S-F reports performance with the cereal classes merged; Section S-G reports the L-TAE-S hyperparameter sensitivity; and Section S-H shows the phenological profiles of the crops. Equation, figure, table, and section labels are prefixed "S".

S-A. Dataset Details, Feature Set, and Model Definitions

Supports main paper Sec. 2–3.

S-A.1. Exploratory Data Analysis

The five crop classes are heavily imbalanced (Fig. S1(a)). Lucerne/medics dominates both regions despite having the smallest mean field area (Fig. S1(b)). Spectral responses overlap substantially across crops (Fig. S2(a)), and wheat and barley are especially similar (Fig. S2(b)), the dominant source of persistent confusion noted in the main paper.

S-A.2. `xr_fresh` Feature Set

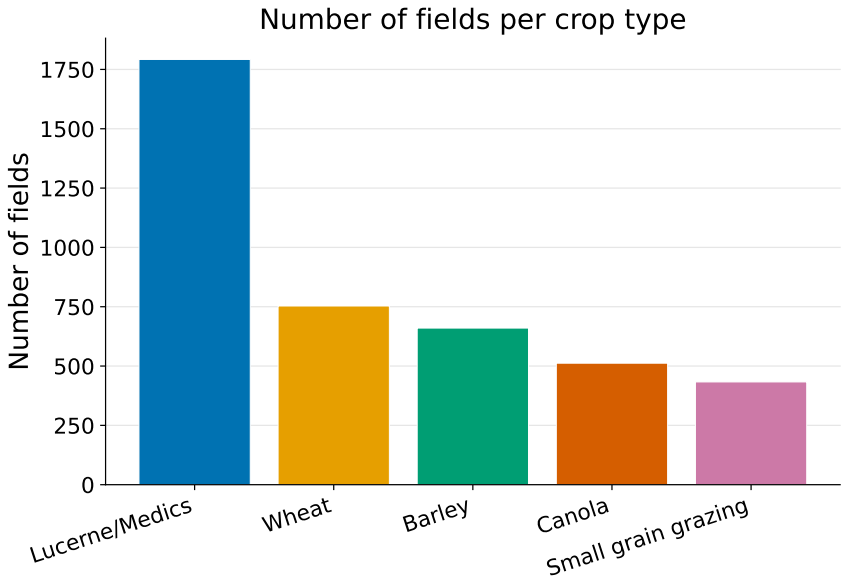
Table S1 lists the time-series features extracted from each pixel's 10-month spectral profile with `xr_fresh` (Mann et al., 2026; Mann, 2025), forming the 174-dimensional per-pixel vector used by the classical learners (Fig. S3 illustrates the extraction).

S-A.3. Baseline Classical Model Definitions

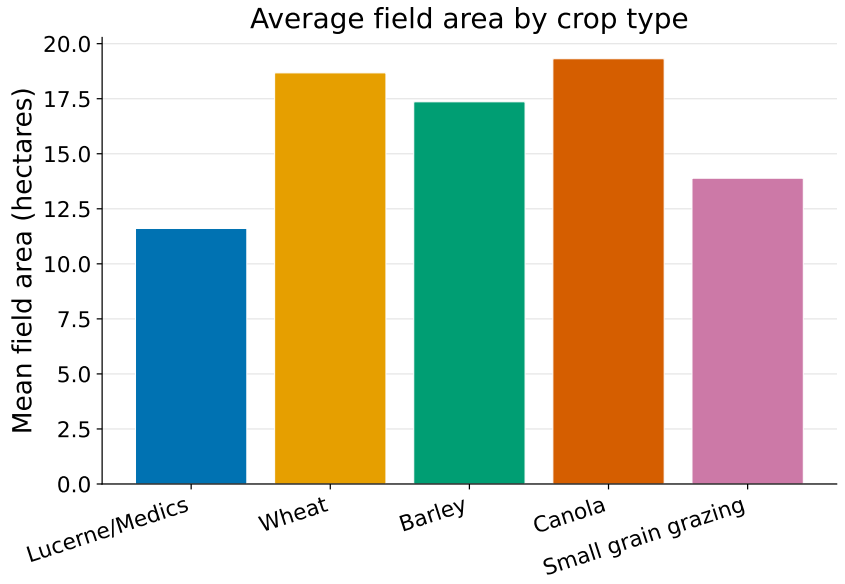
For a feature vector $\mathbf{x}_i$ and class $c \in \{1, \ldots, C\}$, the four baselines are as follows. Logistic regression (multinomial):

$$P(y_i = c \mid \mathbf{x}_i) = \frac{\exp(\mathbf{w}_c^\top \mathbf{x}_i + b_c)}{\sum_{c'=1}^{C} \exp(\mathbf{w}_{c'}^\top \mathbf{x}_i + b_{c'})}. \tag{S1}$$

Random Forest (Breiman, 2001): majority vote over K trees, $\hat{y}_i = \text{mode}\{T_1(\mathbf{x}_i), \ldots, T_K(\mathbf{x}_i)\}$. LightGBM (Ke et al., 2017): additive boosting with rate η, $\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + \eta\, f_t(\mathbf{x}_i)$. XGBoost (Chen and Guestrin, 2016; Friedman, 2001): regularized boosting objective $\mathcal{L} = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k)$ with $\Omega(f_k) = \gamma T_k + \frac{1}{2}\lambda \sum_j w_j^2$. The field-level XGBoost was tuned with Optuna over 150 trials; the top configurations appear in Table S2. Field-level pixel features are aggregated by the mean within each polygon. For the pixel ensemble, RF, XGBoost, and LightGBM probabilities are soft-voted; the field-level stacked ensemble uses RF and XGBoost base learners with a logistic-regression meta-learner, and a confidence-margin rule falls back to pixel-mode voting when the top two field-level class probabilities differ by less than 0.10.

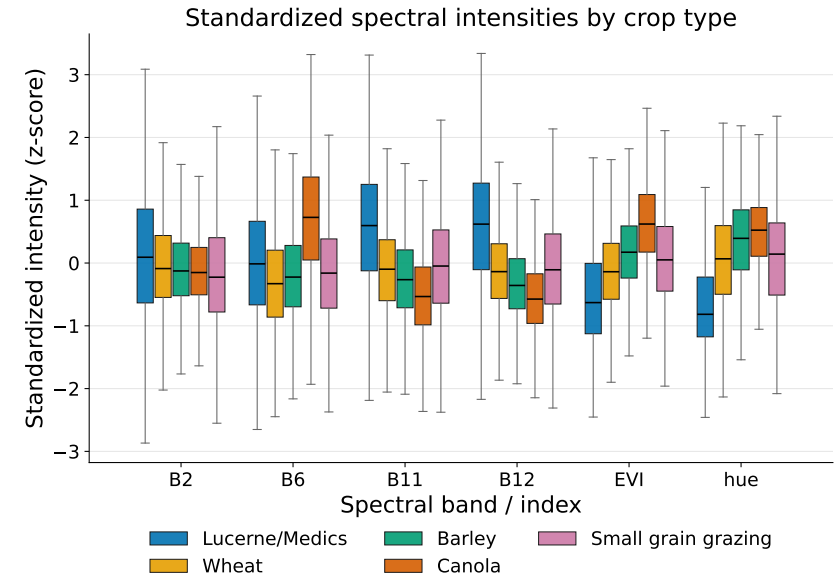


(a)

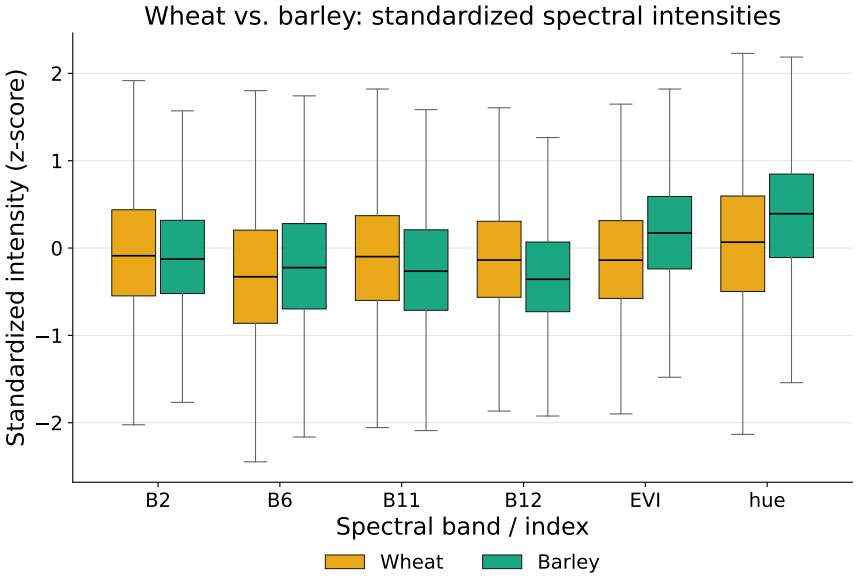


(b)

Figure S1: Crop-field composition of the training and holdout regions. (a) Number of labeled fields per crop type: lucerne/medics dominates (43% of training and 56% of holdout fields), followed by wheat, barley, canola, and small-grain grazing. This heavy class imbalance motivates macro-F1 as the primary metric and the class-imbalance handling used during training. (b) Mean field area by crop type (hectares); lucerne/medics is the most frequent class despite having among the smallest fields.



(a)



(b)

Figure S2: Spectral separability of the crop classes, shown as box plots of standardized (z-score) intensity for each Sentinel-2 band and index (B2, B6, B11, B12, EVI, hue). (a) Distributions for all five crops overlap substantially across every band, so no single feature cleanly separates the classes. (b) Wheat versus barley are especially similar in every band, the dominant source of the persistent wheat–barley confusion seen across all models in the main paper.

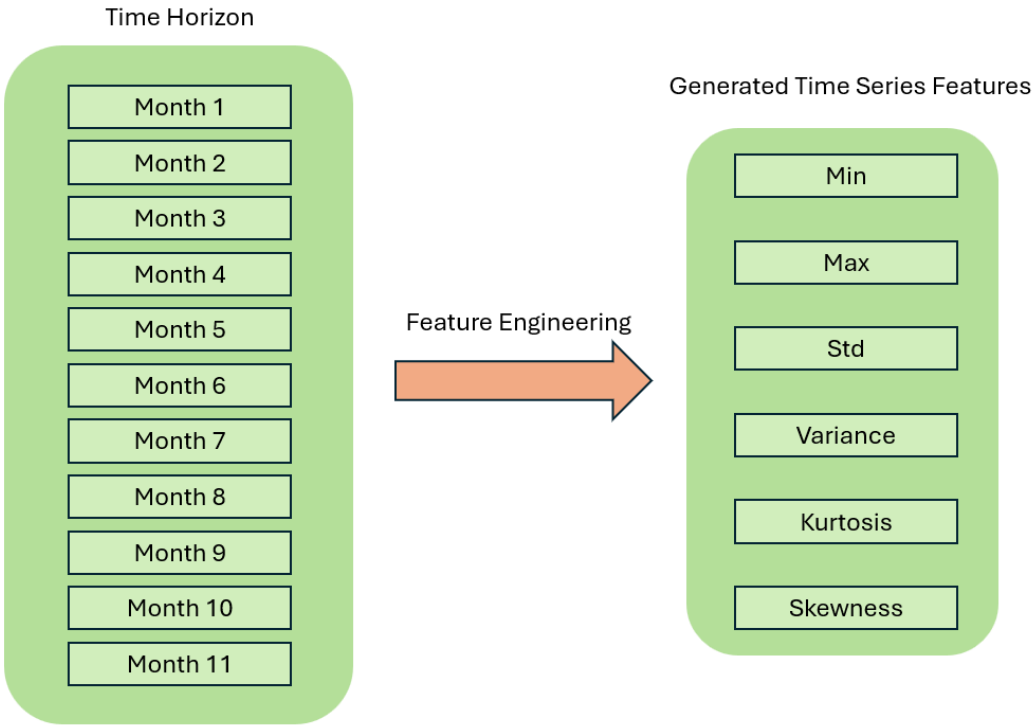


Figure S3: Time-series feature generation with `xr_fresh`. For each pixel, the 10-month profile of every band and index is summarized by the statistical and temporal descriptors of Table S1 (e.g., linear trend, autocorrelation, quantiles, day-of-year of the maximum), producing the 174-dimensional per-pixel feature vector used by the classical learners.

Table S1: Time-series features extracted from each pixel's spectral profile with `xr_fresh`.

Feature	Description
Absolute Energy	Sum of squared values.
Absolute Sum of Changes	Total absolute difference between consecutive measurements.
Autocorrelation (1, 2-month lag)	Correlation with one- and two-month lagged versions.
Count Above Mean	Number of points above the series mean.
Day of Year of Max/Min	Julian day of the maximum and minimum.
Kurtosis	Tail heaviness of the distribution.
Linear Trend	Slope of a least-squares fit.
Longest Strike Above/Below Mean	Longest consecutive run above or below the mean.
Max/Min	Absolute extreme values.
Mean/Median	Measures of central tendency.
Mean Absolute Change/Mean Change	Average magnitude and signed change between points.
Mean Second Derivative	Mean of the series' second differences.
Quantiles (0.05, 0.95)	5th and 95th percentiles.
Ratio Beyond $r\sigma$	Proportion of points more than r s.d. ($r = 1, 2, 3$) from the mean.
Skewness	Asymmetry around the mean.
Standard Deviation/Variance	Spread measures.
Sum of Values	Total sum of observations.
Symmetry Looking	Similarity under series reversal.
Time Series Complexity	Complexity-Invariant Distance metric of irregularity.

Each descriptor is computed independently for every band and index from a pixel's 10-month series; stacked across the six bands/indices (B2, B6, B11, B12, EVI, hue) they form the 174-dimensional per-pixel vector fed to the classical learners. For the Ratio-Beyond features, r is the threshold in standard deviations ($r = 1, 2, 3$).

Table S2: XGBoost hyperparameter search: top 5 Optuna trials with validation accuracy.

Trial	Acc.	Est.	Depth	LR	SubS	ByTree	Gamma	Regα	Regλ	MinCW
32	**76.61%**	1141	13	0.0215	0.5749	0.6986	0.5589	0.00184	0.00259	5
12	76.46%	984	12	0.0145	0.6675	0.6946	0.0799	0.00503	0.00455	7
24	76.40%	1165	11	0.0162	0.7831	0.6403	0.00413	0.00481	0.00478	9
13	76.29%	995	12	0.0277	0.6345	0.6829	0.3999	0.00011	0.00567	6
18	76.29%	915	13	0.0121	0.6784	0.5958	0.1612	0.00491	0.1732	8

Top five of 150 Optuna trials, ranked by in-region validation accuracy (Acc.); trial 32 (bold) was selected for the field-level XGBoost. Est.: number of estimators; Depth: maximum tree depth; LR: learning rate; SubS: row subsample; ByTree: column subsample per tree; Gamma: minimum split-loss reduction; Regα/Regλ: L1/L2 regularization; MinCW: minimum child weight.

S-A.4. Deep Model Definitions

CNN-BiLSTM. An input tensor of shape $(B, 6, T)$ passes a 1D convolution (64 filters, kernel 5) and a bidirectional LSTM (hidden size 64), with the final timestep classified via a softmax head (Fig. S4). Class imbalance is handled with focal loss $\mathcal{L}_{\text{focal}} = -\alpha_t(1 - p_t)^\gamma \log(p_t)$, $\gamma = 2.0$, α_t the inverse class frequency. Five seeds are ensembled by averaging logits.

TabNet. At each of n_{steps} decision steps, an attentive transformer produces a sparse feature mask via *sparsemax*,

$$\begin{aligned} \mathbf{M}[i] &= \text{sparsemax}\big(\mathbf{P}[i-1] \odot h_i(\mathbf{a}[i-1])\big), \\ \mathbf{P}[i] &= \textstyle\prod_{j=1}^{i} \big(\gamma - \mathbf{M}[j]\big), \end{aligned} \tag{S2}$$

with relaxation $\gamma = 1.5$; the masked features are handled by a feature transformer. Unlike an ℓ_1 norm, which is fixed at 1 on the simplex, TabNet's mask-entropy penalty $\lambda_{\text{sparse}} \sum_i \sum_j -M_j[i] \log M_j[i]$ ($\lambda_{\text{sparse}} = 10^{-3}$, library default) varies with mask concentration and so adds genuine sparsity pressure. We use $n_d = n_a = 64$, 5 decision steps, batch size 1024, and ensemble five seeds by averaging softmax outputs. The sparsemax

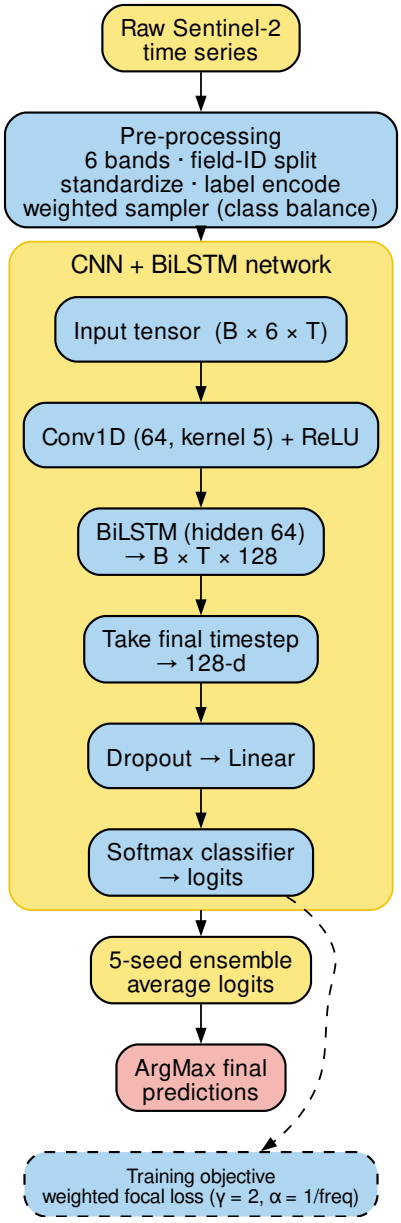


Figure S4: Architecture of the CNN-BiLSTM model. The $(B, 6, T)$ input (six bands/indices over T months) passes a 1D convolution (64 filters, kernel 5) for temporal feature extraction, then a bidirectional LSTM (hidden size 64) whose final time step is classified by a softmax head. Training uses weighted focal loss to counter class imbalance, and five seeds are ensembled by averaging logits.

masks are the mechanism the main paper's L-TAE-S grafts onto a temporal-attention backbone (Arik and Pfister, 2021).

L-TAE. Given an embedded sequence $\{\mathbf{z}_t\}_{t=1}^{T}$, a learned master query attends over time; for a single head

$$\mathbf{o} = \sum_{t=1}^{T} \alpha_t \, \mathbf{v}_t, \qquad \alpha_t = \frac{\exp(\mathbf{q}^\top \mathbf{k}_t/\sqrt{d})}{\sum_{t'} \exp(\mathbf{q}^\top \mathbf{k}_{t'}/\sqrt{d})}, \tag{S3}$$

with channel grouping keeping the encoder lightweight (Garnot and Landrieu, 2020). Evaluated at pixel and field level as a five-seed ensemble.

Transformer encoder. After the same band embedding and month-aware positional encoding as L-TAE, the sequence passes three pre-norm multi-head self-attention layers ($d_{\text{model}} = 128$, 8 heads, feed-forward width 256), is mean-pooled, and classified (Vaswani et al., 2017; Rußwurm and Körner, 2020). Unlike L-TAE-S it has no feature-sparsity mechanism, serving as the capacity-versus-sparsity control. Five-seed ensemble under the same weighted-focal objective.

TempCNN. Stacked 1D convolutions along the temporal axis with batch normalization and dropout, followed by a softmax head (Pelletier et al., 2019); pixel and field variants, five seeds.

Patch models. Each field is tiled into 100 m patches ($\sim 10 \times 10$ pixels) resampled to 128×128 (Fig. S5). The multi-channel 2D CNN treats each band×month as a channel ($C = 60$) through two convolutional blocks; the 3D CNN treats each patch as a spatiotemporal volume ($T = 10$, $C = 6$) through three 3D-convolutional blocks with a weighted focal loss and early stopping on validation macro-F1 (Fig. S6); the 2D CNN uses unweighted cross-entropy for a fixed 20 epochs, with no dropout. After upsampling, the receptive field of the final convolution spans ∼10 (2D) and ∼18 (3D) upsampled pixels, about one to two native 10 m pixels. Both are five-seed ensembles aggregated to a field label by majority vote.

S-B. Performance Metric Definitions

Supports main paper Sec. 3. Let N be the number of samples, C the classes, and n_c the support of class c. Cohen's Kappa is $\kappa = (p_o - p_e)/(1 - p_e)$ with p_o the observed agreement and $p_e = \sum_c (n_{c,\text{true}}/N)(n_{c,\text{pred}}/N)$. Macro-F1 (primary) and weighted F1 are

$$\text{F1}_m = \frac{1}{C} \sum_{c=1}^{C} \text{F1}_c, \quad \text{F1}_\text{w} = \sum_{c=1}^{C} \frac{n_c}{N} \, \text{F1}_c, \tag{S4}$$

with $\text{F1}_c = 2\,\text{Prec}_c\,\text{Rec}_c/(\text{Prec}_c + \text{Rec}_c)$. Cross-entropy (Xent) is the AI4FoodSecurity field-level scoring metric (AI4EO, 2021); because submissions are hard labels, it is evaluated on the one-hot predicted label clipped to $[\epsilon, 1-\epsilon]$, $\epsilon = 10^{-7}$:

$$\begin{aligned} \text{Xent} &= -\frac{1}{N} \sum_{i=1}^{N} \sum_{c=1}^{C} y_{i,c} \, \ln \hat{q}_{i,c}, \\ \hat{q}_{i,c} &= \text{clip}\big(\mathbb{1}[\hat{y}_i = c],\, \epsilon,\, 1-\epsilon\big). \end{aligned} \tag{S5}$$

S-C. Training Cost

Supports main paper Sec. 3 and the Discussion. Table S3 reports wall-clock training time for a single run of each model on one workstation (RTX 3080 Ti). The robust models that transfer best (the `xr_fresh` tree ensembles and the field-level L-TAE-S) are among the cheapest to train, while the most expensive (the patch CNNs and the pixel-level Transformer) do not lead the holdout ranking.

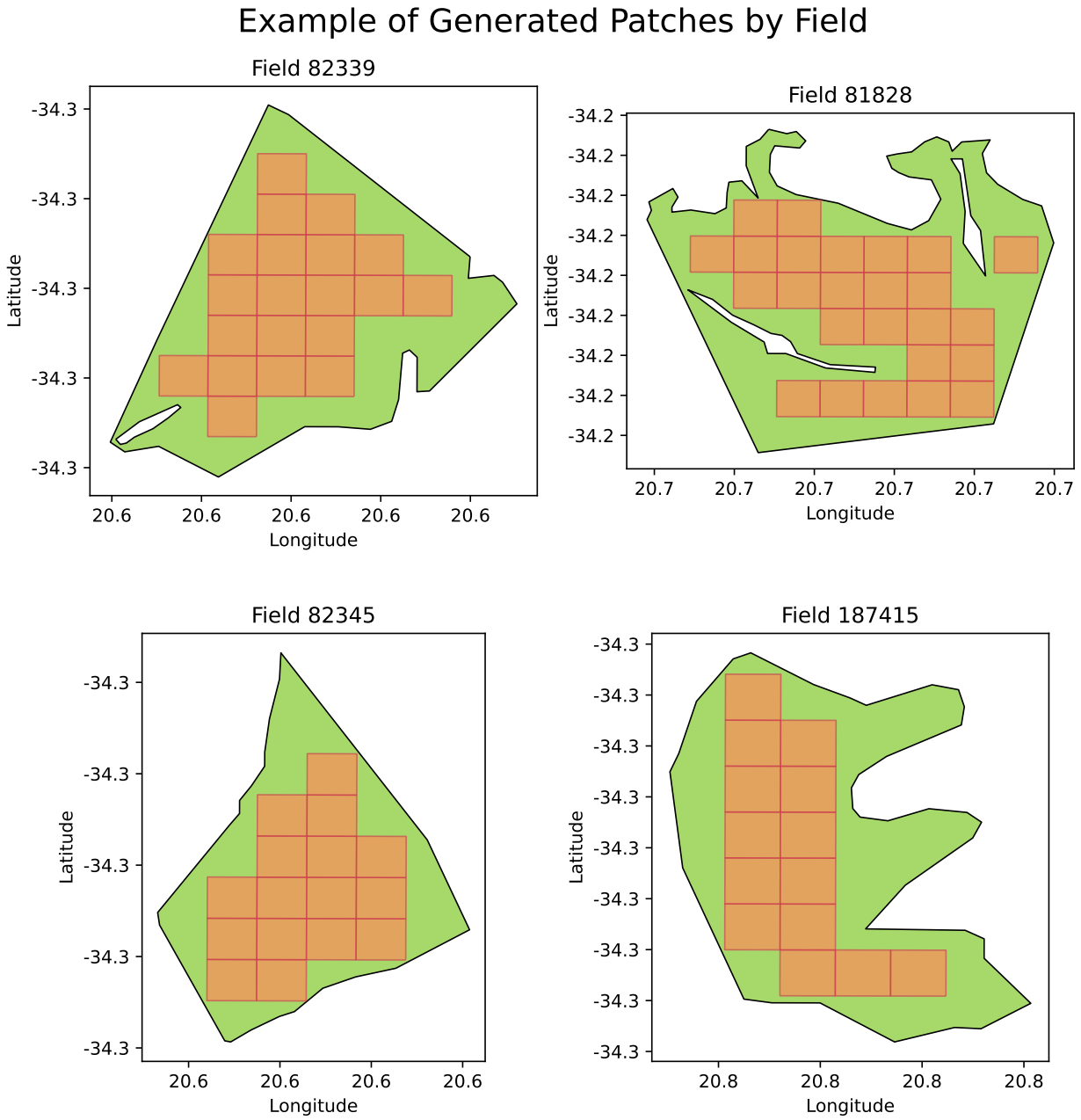


Figure S5: Patch generation per field. Each field is tiled into 100 m patches (only $\sim 10 \times 10$ pixels at the 10 m Sentinel-2 resolution) and resampled to 128×128. These patches are the input to the patch-based 2D and 3D CNNs, whose per-patch predictions are later aggregated to a field label by majority vote.

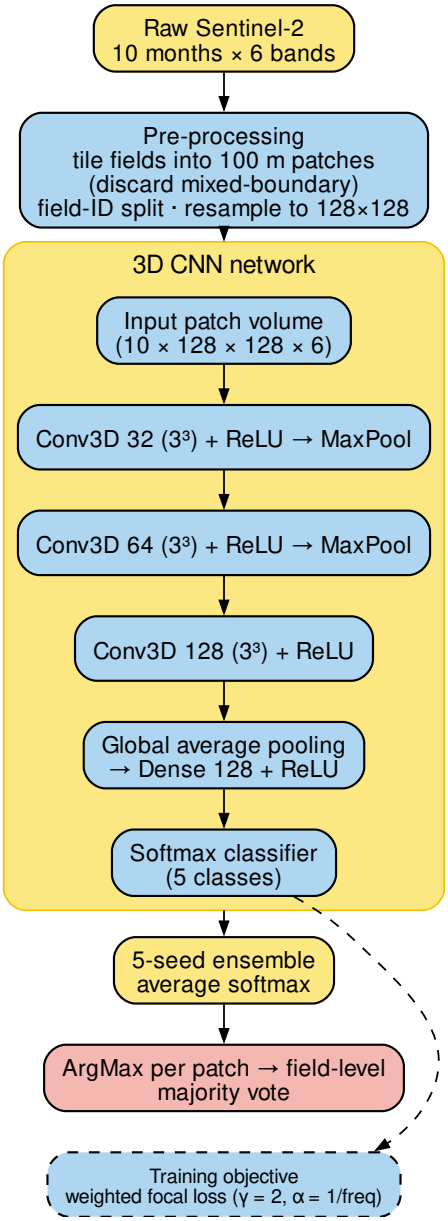


Figure S6: Patch-level 3D CNN pipeline: each field is tiled into 100 m patches resampled to 128 × 128; a five-seed ensemble averages softmax outputs, and per-patch predictions are aggregated to a field label by majority vote.

Table S3: Per-seed training time for each model.

Model	Level	Features	Time / run
Classical, field-level			
XGBoost	field	`xr_fresh`	12 s
LightGBM	field	`xr_fresh`	31 s
Voting	field	`xr_fresh`	109 s
Stacking	field	`xr_fresh`	109 s
SMOTE Stacked	field	`xr_fresh`	99 s
Classical, pixel-level			
XGBoost	pixel	`xr_fresh`	31 s
LightGBM	pixel	`xr_fresh`	31 s
Logistic Regression	pixel	`xr_fresh`	8.0 min
Random Forest	pixel	`xr_fresh`	73 min
Deep temporal, field-level (raw sequences)			
TabNet (`xr_fresh`)	field	`xr_fresh`	34 s
TabNet (raw, field-avg)	field	raw	39 s
CNN-BiLSTM	field	raw	53 s
L-TAE	field	raw	61 s
TempCNN	field	raw	64 s
L-TAE-S	field	raw	18 s
TimesNet	field	raw	26 s
Deep temporal, pixel-level (raw sequences)			
CNN-BiLSTM	pixel	raw	10 min
L-TAE-S	pixel	raw	24 min
TempCNN	pixel	raw	29 min
TimesNet	pixel	raw	30 min
L-TAE	pixel	raw	38 min
TabNet (raw)	pixel	raw	61 min‡
Transformer	pixel	raw	97 min
Random-convolution classifier (raw sequences)			
MiniRocket	field	raw	30 s
MiniRocket	pixel	raw	29 min§
Deep patch-based (raw sequences)			
3D CNN	patch	raw	22 min
Multi-Channel 2D CNN	patch	raw	38 min

Deep-model times are the mean per seed of a 5-seed ensemble (total wall-clock divided by five); classical models are a single best-configuration fit, excluding the one-time Optuna search. ‡Measured on the 75% training fraction, the best-transferring TabNet pixel configuration. §Trained on at most 50 pixels per field (Sec. S-I). TimesNet and MiniRocket times are measured per seed.

S-D. Ablation Studies

Supports main paper Sec. 5.2 ("Two Robustness Levers"). We attribute the holdout ranking primarily to two levers, built-in sparse feature selection and field-level aggregation, and show that ensemble size and the imbalance-handling objective are second-order: they shift absolute scores but never reorder the robust and fragile model groups.

(i) Field-level aggregation (main Table 2) narrows the transfer gap for dense temporal nets (L-TAE $-0.24 \rightarrow -0.07$; L-TAE-S $-0.17 \rightarrow -0.04$), raising holdout macro-F1 for L-TAE but not for TempCNN, CNN-BiLSTM, TimesNet, or MiniRocket, and harms TabNet ($0.60 \rightarrow 0.43$). *(ii) Sparse selection versus attention capacity*: a plain Transformer with far greater capacity improves on L-TAE (OOR $F1_m$ 0.58 vs. 0.54), but grafting a sparse channel gate onto L-TAE (L-TAE-S) lifts transfer further, to 0.60. *(iii) Resampling*: the *Stacking* and *SMOTE Stacked* ensembles differ only in SMOTETomek resampling; resampling widens the gap ($-0.11 \rightarrow -0.21$) and lowers holdout $F1_m$ ($0.53 \rightarrow 0.49$).

(iv) Ensemble size (Table S4): seed-ensembling gives a small, monotone lift but does not change the ordering: a single-seed L-TAE-S ($F1_m = 0.56$) already exceeds the five-seed CNN-BiLSTM and TabNet field ensembles. *(v) Imbalance-handling objective* (Table S5): the robust L-TAE-S is almost invariant across five loss/sampler settings (0.59–0.60), whereas the fragile CNN-BiLSTM is sensitive (0.41–0.56); even CNN-

Table S4: Ablation (iv): out-of-region transfer holdout performance versus seed-ensemble size.

Model	**Ensemble**	**F1**$_m$	κ
TabNet (field)	1 seed	0.31 ± 0.02	0.18 ± 0.03
	3 seeds	0.39	0.29
	5 seeds	0.43	0.34
L-TAE-S (field)	1 seed	0.56 ± 0.02	0.47 ± 0.02
	3 seeds	0.58	0.50
	5 seeds	0.60	0.52
CNN-BiLSTM (field)	1 seed	0.48 ± 0.02	0.31 ± 0.02
	3 seeds	0.52	0.34
	5 seeds	0.51	0.33

By re-inference of saved per-seed checkpoints. Single-seed rows report mean±s.d. across five seeds.

Table S5: Ablation (v): effect of the imbalance-handling objective and resampling on out-of-region transfer holdout performance.

Model	**Loss**	**Resample**	**F1**$_m$	κ
TabNet (field)	Weighted focal (γ=2)	no	0.43	0.34
	Weighted CE (γ=0)	no	0.42	0.33
	Plain CE	no	0.35	0.25
	Weighted focal (γ=2)	yes	0.40	0.23
	Plain CE	yes	0.46	0.37
L-TAE-S (field)	Weighted focal (γ=2)	no	0.60	0.52
	Weighted CE (γ=0)	no	0.60	0.50
	Plain CE	no	0.59	0.50
	Weighted focal (γ=2)	yes	0.60	0.49
	Plain CE	yes	0.60	0.50
CNN-BiLSTM (field)	Weighted focal (γ=2)	no	0.51	0.34
	Weighted CE (γ=0)	no	0.49	0.34
	Plain CE	no	0.41	0.30
	Weighted focal (γ=2)	yes	0.56	0.39
	Plain CE	yes	0.50	0.34

Field-level, 5-seed ensembles. "Resample" denotes class-balanced oversampling.

BiLSTM's best setting (0.56) stays below L-TAE-S's worst (0.59), so the objective never reorders the two.

S-E. Spatial Distribution of Holdout Errors

Supports main paper Sec. 4.3. Figure S7 shows the correct and incorrectly classified holdout fields for the five representative models, with contours at 5, 10, and 15 km from the nearest training field. The errors are spatially clustered rather than changing uniformly. Nearer the training tiles (< 5 km, 521 fields) all five models classify about 70% of fields correctly (69 to 72%). In a south-central zone 8 to 16 km from the training tiles (398 fields), L-TAE and CNN-BiLSTM fall to 36% and 30%, whereas TabNet (70%), L-TAE-S (67%), and field-level XGBoost (63%) remain close to their tile-wide accuracy. The zone is contiguous, and accuracy does not decline steadily with distance, so the pattern reflects local conditions within the tile rather than distance from the training region. Hence, the dense temporal networks are the models most affected by these local conditions.

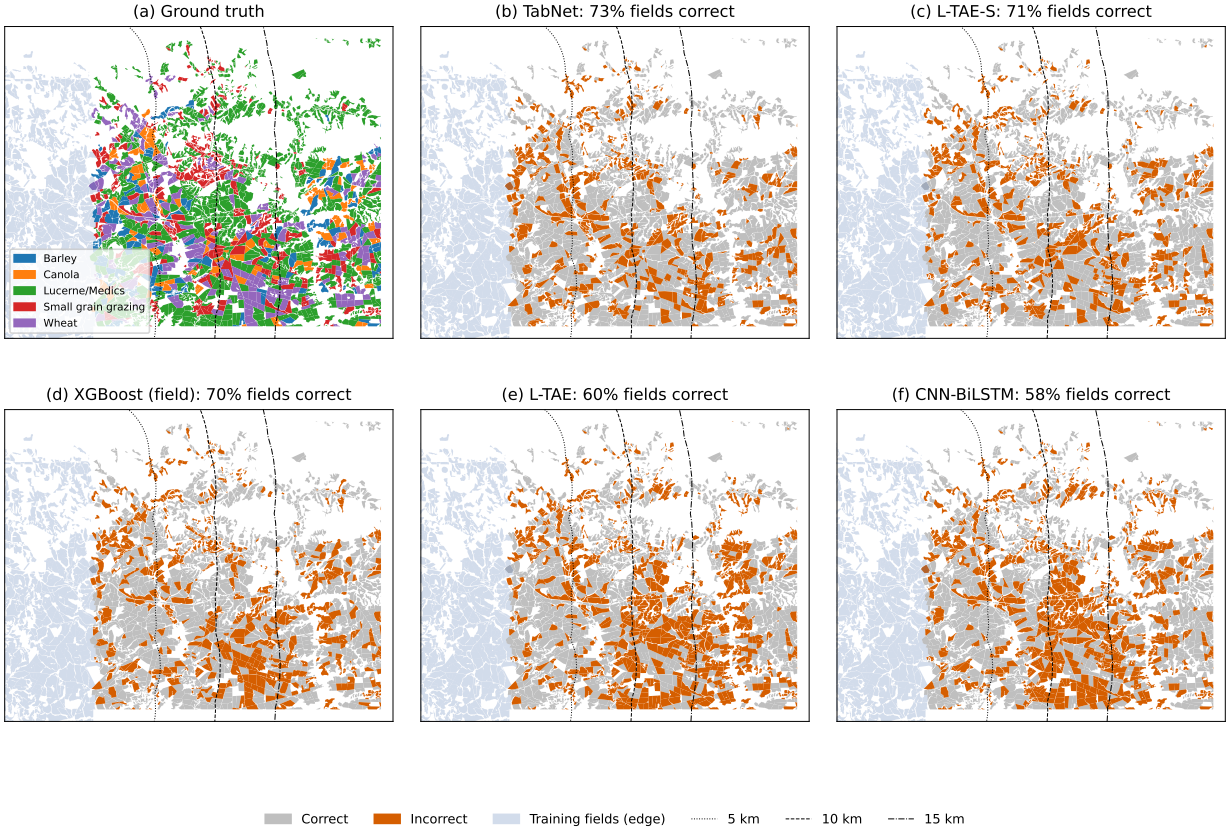


Figure S7: Spatial distribution of holdout errors. (a) Ground-truth crop types. (b)–(f) Correct (grey) and incorrect (orange) fields for TabNet, L-TAE-S, field-level XGBoost, L-TAE, and CNN-BiLSTM; panel titles give the share of fields classified correctly. Light blue shows the edge of the training region; dotted, dashed, and dash-dot lines mark 5, 10, and 15 km from the nearest training field.

S-F. Merged Cereal Classes

Supports main paper Sec. 5.3. Table S6 re-scores every model's holdout predictions after merging barley, wheat, and small-grain grazing into a single cereal class. Accuracy rises by 0.06 to 0.16 for every model, and the sparse models remain at the top while the dense pixel networks stay at the bottom.

Table S6: Out-of-region performance with the cereals merged into one class.

		Macro-F1		**Accuracy**		**Cereal**
Model	**Level**	**5 cl.**	**3 cl.**	**5 cl.**	**3 cl.**	**F1**
TabNet	pixel	**0.60**	0.82	**0.73**	**0.84**	0.76
L-TAE-S	field	**0.60**	**0.83**	0.70	0.83	0.77
L-TAE-S	pixel	**0.60**	0.80	0.71	0.83	**0.78**
L-TAE	field	0.59	0.78	0.66	0.78	0.74
Transformer	pixel	0.58	0.77	0.69	0.82	0.76
TempCNN	pixel	0.57	0.76	0.69	0.81	0.75
TimesNet†	pixel	0.57	0.74	0.67	0.79	0.74
XGBoost	field	0.56	0.77	0.70	0.81	0.73
LightGBM	pixel	0.56	0.78	0.70	0.80	0.69
Logistic Regression	pixel	0.56	0.76	0.69	0.80	0.72
LightGBM	field	0.56	0.76	0.69	0.79	0.69
Voting	pixel	0.56	0.80	0.70	0.81	0.71
Random Forest	pixel	0.55	0.78	0.69	0.79	0.69
XGBoost	pixel	0.54	0.77	0.69	0.81	0.74
L-TAE	pixel	0.54	0.71	0.60	0.73	0.70
TempCNN	field	0.54	0.73	0.62	0.77	0.73
Stacking	field	0.53	0.74	0.67	0.80	0.76
TimesNet†	field	0.52	0.67	0.57	0.70	0.66
CNN-BiLSTM	pixel	0.51	0.71	0.58	0.73	0.70
CNN-BiLSTM	field	0.51	0.69	0.51	0.67	0.67
3D CNN	patch	0.50	0.67	0.51	0.64	0.63
MiniRocket†	pixel	0.50	0.66	0.52	0.66	0.66
SMOTE Stacked	field	0.49	0.68	0.67	0.76	0.65
MiniRocket†	field	0.48	0.66	0.51	0.66	0.65
Multi-Channel 2D CNN	patch	0.45	0.68	0.53	0.68	0.64
TabNet (raw, field-avg)	field	0.43	0.64	0.54	0.70	0.67
TabNet (`xr_fresh`)	field	0.37	0.54	0.62	0.68	0.46

Models in the order of main Table 1. 5 cl.: the five published classes; 3 cl.: barley, wheat, and small-grain grazing merged into one cereal class, giving cereal, canola, and lucerne/medics. Classes are merged in each model's existing holdout confusion matrix, without retraining. Cereal F1 is the F1 of the merged class. Part of the macro-F1 increase is mechanical, because there are fewer classes and the hardest ones are combined, so accuracy and cereal F1 are the more comparable measures. †Recent time-series classifiers (Sec. S-I). Bold marks the best value per column.

S-G. L-TAE-S Hyperparameter Sensitivity

Supports main paper Secs. 3.2.1 and 5.1. Table S7 reports out-of-region macro-F1 for field-level L-TAE-S over a grid of the relaxation coefficient γ and the number of attention heads (key dimension fixed at 8), each trained as a five-seed ensemble with the same splits and protocol as the published model, together with an explicit mask-entropy sparsity penalty and two pixel-level settings. Retraining the published configuration reproduces the published predictions exactly. Twenty-two of the 23 field-level settings fall between 0.58 and 0.62, a range comparable to the seed-to-seed variation (single-seed standard deviation 0.004 to 0.028), and the published setting lies mid-range. Fewer heads transfer slightly better (mean 0.61 with 4 heads, 0.59 with 32), γ has no consistent effect, and the entropy penalty leaves accuracy unchanged, so sparsemax and the relaxation prior already provide the sparsity. The two pixel-level settings bracket the published pixel model (0.60): 4 heads with $\gamma = 2.0$ reaches 0.61 and 32 heads with $\gamma = 1.0$ reaches 0.57. In paired bootstrap tests both still improve on the dense L-TAE (Δ macro-F1 +0.07 and +0.03; $\Delta\kappa$ +0.15 and +0.11, all significant), so the gain from the sparse gate does not depend on the chosen hyperparameters, although its size does.

Table S7: L-TAE-S out-of-region macro-F1 across hyperparameter settings.

(a) Field level, attention heads	**4**	**8**	**16**	**32**
$\gamma = 1.0$	0.61	0.61	0.59	0.58
$\gamma = 1.2$	0.60	0.61	0.59	0.60
$\gamma = 1.5$	0.61	0.60	**0.60**	0.58
$\gamma = 2.0$	0.61	0.61	0.60	0.59
$\gamma = 3.0$	0.61	0.59	0.58	0.16[†]
(b) Field level, explicit sparsity penalty				
Entropy penalty, $\lambda = 10^{-3}, 10^{-2}, 10^{-1}$ ($\gamma = 1.5$, 16 heads)	0.60, 0.59, 0.61			
(c) Pixel level				
$\gamma = 2.0$, 4 heads	0.61 (in-region 0.76)			
$\gamma = 1.5$, 16 heads (published)	**0.60** (in-region 0.77)			
$\gamma = 1.0$, 32 heads	0.57 (in-region 0.78)			

Five-seed ensembles, full training set, same splits and protocol as main Table 1; bold marks the published configuration. Field-level dense L-TAE scores 0.59 and pixel-level L-TAE 0.54. [†]Training failed on all five seeds (validation macro-F1 stayed near 0.2): the relaxation prior is multiplied by up to γ at each head and reaches $\approx 3^{31}$, overwhelming the learned gate.

S-H. Crop Phenology

Supports main paper Secs. 2 and 5.3. Figure S8 shows field-level monthly profiles of EVI, SWIR (B11, B12), and hue for each crop over the ten months used by the models. Lucerne/medics, a pasture class, has the most distinct profile: a weak winter peak (median EVI about 0.32, against 0.45 to 0.51 for the annual crops) and higher SWIR reflectance in August and September. Canola greens up earliest (median EVI near 0.50 already in July) and has the lowest mid-season SWIR. The three cereals follow nearly the same curve in every band; small-grain grazing is a cereal such as oats sown as winter fodder for livestock rather than harvested for grain, so it is often the same kind of plant as wheat or barley. The largest standardized difference (Cohen's d) between two cereals in any band and month is 0.73 (barley versus small-grain grazing) and only 0.50 for wheat versus barley, against 1.1 to 2.3 for every pair involving lucerne/medics or canola. Time-series features alone therefore separate lucerne/medics and canola well but not the cereals, consistent with the per-class results (main Sec. 4.3) and the merged-cereal analysis (Sec. S-F).

The holdout season is also later. July EVI of the annual crops is 0.11 to 0.21 lower on the holdout tile than in the training region, and October EVI is slightly higher, whereas lucerne/medics changes little. This shift is larger than the differences among the cereals, so it adds to their confusion under transfer. May and June, when the winter crops are sown and emerge, are excluded because of cloud cover, which removes early-season information that could help separate them.

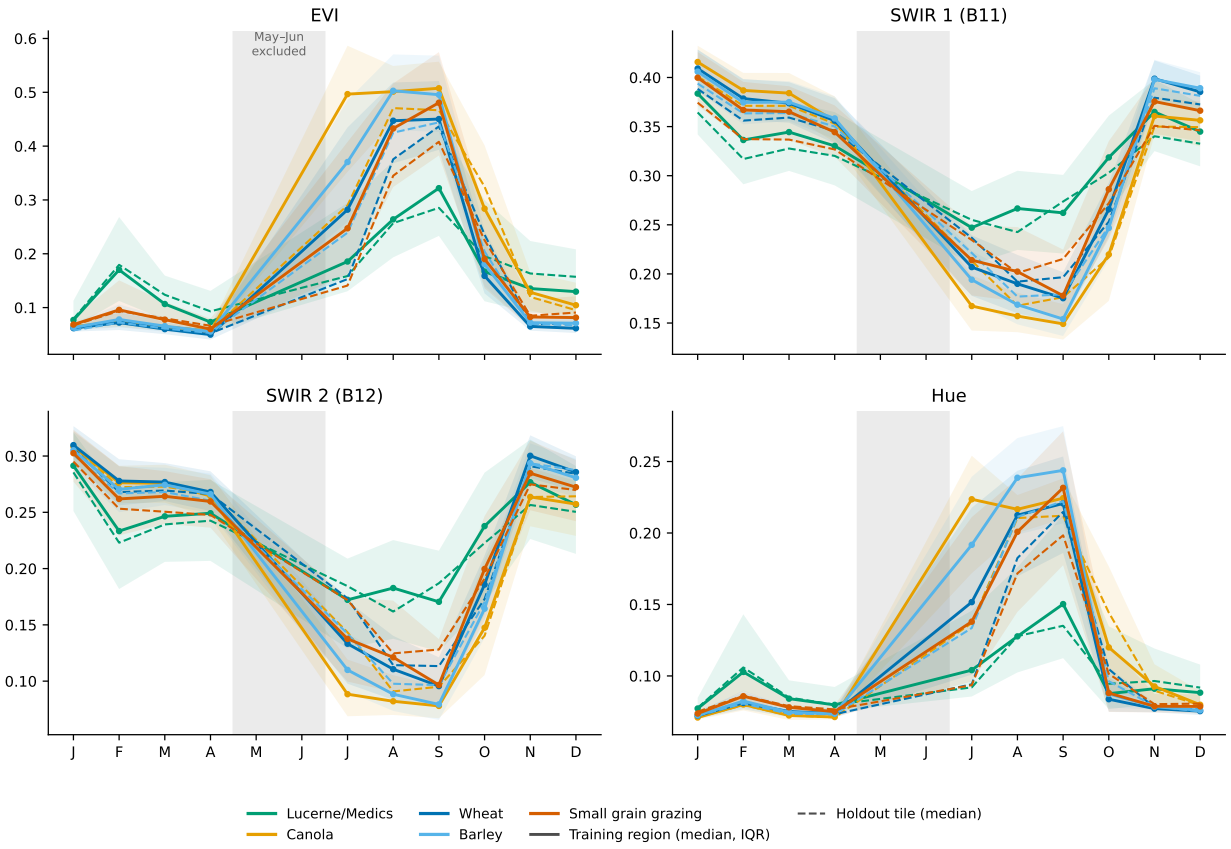


Figure S8: Field-level monthly profiles of EVI, SWIR 1 (B11), SWIR 2 (B12), and hue for the five crops. Solid lines and shading show the median and interquartile range across training-region fields; dashed lines show the holdout-tile median. The grey band marks May and June, excluded because of cloud cover.

S-I. Recent Time-Series Baselines and Sparse Selection on MiniRocket

Supports main paper Secs. 3, 4.1, 4.2, 5.1, and 5.2. This section describes the two recent time-series classifiers included in main Tables 1 and 2 and a follow-up experiment on MiniRocket. TimesNet (Wu et al., 2023) is a high-capacity network that reshapes a series into two dimensions by its dominant periods. MiniRocket (Dempster et al., 2021) passes about 10,000 random convolution features to a linear classifier and ranks among the most accurate general time-series classifiers. For a fair comparison we use the same 60 inputs, field-wise split, scaler, five seeds, field aggregation, and holdout scoring as the other models, and we checked that their inputs are identical. TimesNet uses the L-TAE-S training settings. MiniRocket uses a logistic-regression head with the penalty chosen on the validation split; its standard ridge head performs no better (holdout $F1_m$ 0.49 pixel, 0.45 field). At the pixel level, MiniRocket trains on at most 50 pixels per field because of memory limitations. We did not run a pretrained encoder such as Presto, because it needs Sentinel-2 bands that are not in the processed benchmark.

We find that neither model transfers as well as the sparse learners (main Table 1). On the same holdout fields, TimesNet trails L-TAE-S by 0.03 at pixel level (paired 95% CI $[0.01, 0.06]$) and 0.08 at field level ($[0.06, 0.11]$), and MiniRocket trails it by 0.10 and 0.12. TimesNet behaves like the other dense temporal networks: high in-region accuracy with a large performance drop on the holdout. MiniRocket falls below logistic regression on the raw inputs. For both, field aggregation lowers holdout accuracy.

To test whether sparse feature selection helps a second backbone, we kept each seed's MiniRocket features fixed and changed only the classifier (Table S8). The sparse head is the L-TAE-S gate: 16 heads that each keep about two features per sample. The dense control is the identical network with softmax in place of sparsemax. At the pixel level the sparse gate improves holdout $F1_m$ by 0.058 over the dense control (paired 95% CI $[0.040, 0.075]$; $\Delta\kappa = 0.10$), close to the 0.06 gain the gate gives L-TAE, while the dense gate adds nothing over the linear head (-0.013, $[-0.034, 0.007]$). Removing features globally with an L1 penalty does not help ($+0.008$), so the gain comes from choosing different features for each sample. At the field level the sparse and dense gates do not differ (-0.005, $[-0.027, 0.016]$), matching the non-significant field-level gain of L-TAE-S over L-TAE. Even with the gate, MiniRocket stays below L-TAE-S (-0.054 pixel, $[-0.080, -0.028]$).

Table S8: Out-of-region macro-F1 of different classifiers on the same MiniRocket features.

Classifier on MiniRocket features	**Pixel**	**Field**
Linear, L2 logistic (main Table 1)	0.50	0.48
Linear, ridge (standard MiniRocket head)	0.49	0.45
Linear, L1 logistic (global selection)	0.51	0.49
Dense gate (softmax)	0.49	0.50
Sparse gate (sparsemax, as in L-TAE-S)	**0.55**	**0.50**
L-TAE-S (reference)	0.60	0.60

All rows use the same per-seed MiniRocket features and training rows; only the classifier differs. The two gated networks are identical except for sparsemax versus softmax in the gate and are trained with the L-TAE-S settings. The L1 head keeps on average 4,125 (pixel) and 1,175 (field) of the 9,996 features. Paired comparisons are given in the text. Bold marks the best MiniRocket value per column.

References

AI4EO, 2021. AI4FoodSecurity challenge (South Africa). European Space Agency (ESA), in partnership with Planet, TUM, DLR and Radiant Earth, https://platform.ai4eo.eu/ai4food-security-south-africa.

Arik, S.Ö., Pfister, T., 2021. TabNet: Attentive interpretable tabular learning. Proceedings of the AAAI Conference on Artificial Intelligence 35, 6679–6687. doi:10.1609/aaai.v35i8.16826.

Breiman, L., 2001. Random forests. Machine Learning 45, 5–32. doi:10.1023/A:1010933404324.

Chen, T., Guestrin, C., 2016. XGBoost: A scalable tree boosting system, in: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 785–794. doi:10.1145/2939672.2939785.

Dempster, A., Schmidt, D.F., Webb, G.I., 2021. MiniRocket: A very fast (almost) deterministic transform for time series classification, in: Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, pp. 248–257. doi:10.1145/3447548.3467231.

Friedman, J.H., 2001. Greedy function approximation: A gradient boosting machine. Annals of Statistics 29, 1189–1232. doi:10.1214/aos/1013203451.

Garnot, V.S.F., Landrieu, L., 2020. Lightweight temporal self-attention for classifying satellite images time series, in: Advanced Analytics and Learning on Temporal Data (AALTD 2020), ECML PKDD Workshops, Springer. pp. 171–181. doi:10.1007/978-3-030-65742-0_12.

Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., Liu, T.Y., 2017. LightGBM: A highly efficient gradient boosting decision tree, in: Advances in Neural Information Processing Systems (NeurIPS).

Mann, M.L., 2025. xr_fresh: Automated time series feature extraction for remote sensing and gridded data. Journal of Open Source Software 10, 9009. doi:10.21105/joss.09009.

Mann, M.L., Colson, L., Nealon, R., Engstrom, R., Nakacwa, S., 2026. Lite learning: Efficient crop classification in Tanzania using feature extraction with machine learning and crowd sourcing. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing 19, 24506–24519. doi:10.1109/jstars.2026.3708028.

Pelletier, C., Webb, G.I., Petitjean, F., 2019. Temporal convolutional neural network for the classification of satellite image time series. Remote Sensing 11, 523. doi:10.3390/rs11050523.

Rußwurm, M., Körner, M., 2020. Self-attention for raw optical satellite time series classification. ISPRS Journal of Photogrammetry and Remote Sensing 169, 421–435. doi:10.1016/j.isprsjprs.2020.06.006.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł., Polosukhin, I., 2017. Attention is all you need, in: Advances in Neural Information Processing Systems (NeurIPS), pp. 5998–6008.

Wu, H., Hu, T., Liu, Y., Zhou, H., Wang, J., Long, M., 2023. TimesNet: Temporal 2d-variation modeling for general time series analysis, in: International Conference on Learning Representations (ICLR). URL: https://openreview.net/forum?id=ju_Uqw384Oq.